



CIVICEYE AI

YCS 2026

AI-Powered Civic Issue Reporting Platform

Sri Lanka

Competition Prototype · YCS 2026

CivicEye AI · YCS 2026 · Sri Lanka

CivicEye AI is an AI-assisted civic issue reporting and monitoring web application developed as a competition prototype for YCS 2026. The platform is designed to help citizens report common public issues such as road damage, garbage, water leaks, and faulty street lights through a structured digital workflow.

Citizens can submit issue information including an image, issue type, description, and location. The system then performs AI-assisted analysis using programmed rules and keyword matching to support issue categorization, priority assessment, and authority recommendation.

The platform also demonstrates duplicate detection, citizen notifications, report tracking, resolution verification, and an administrator dashboard.

Prototype Scope: CivicEye AI is a prototype and is not currently connected to an official Municipal Council, government authority, emergency service, or production government database.

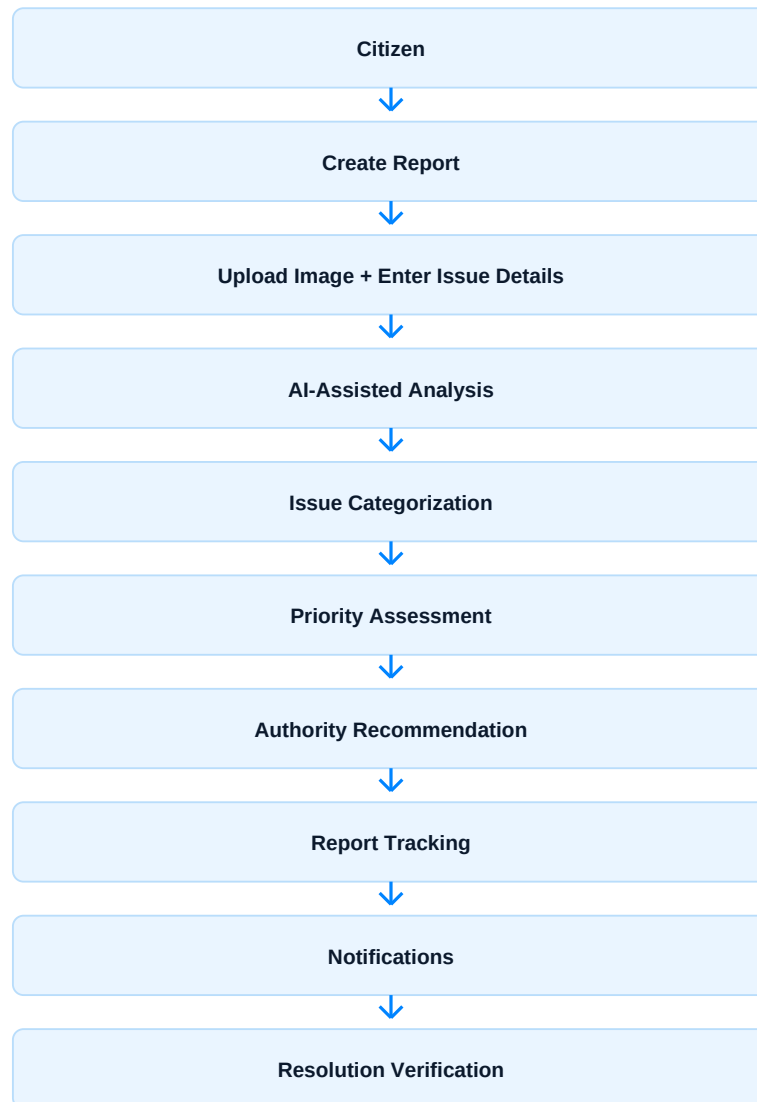
Public issues such as damaged roads, garbage accumulation, water leaks, and faulty street lights require an organized way for citizens to report them and for responsible authorities to monitor them.

Traditional reporting processes may involve difficulties such as:

- Unclear reporting channels
- Incomplete complaint information
- Difficulty identifying the relevant authority
- Repeated or duplicate complaints
- Limited visibility of complaint progress
- Difficulty identifying higher-priority issues
- Language barriers for some citizens

CivicEye AI addresses these challenges through a structured digital reporting workflow that combines citizen input, AI-assisted analysis, location information, priority assessment, authority recommendation, duplicate detection, notifications, and report tracking.

CivicEye AI provides a single web-based platform where citizens can create and monitor civic issue reports.
THE GENERAL WORKFLOW IS:



The system is designed to make the reporting process more structured while providing administrators with a dashboard for monitoring submitted reports.

4.1 AI-Assisted Issue Categorization

The system analyzes the submitted issue information using programmed rules and keyword matching to determine the likely issue category.

Supported example categories include: Road Damage · Garbage · Water Leak · Faulty Street Light

4.2 Image-Based Reporting

Citizens can upload an image related to the reported issue. The image provides supporting visual information for the submitted complaint.

The current prototype does not use a trained computer-vision model for image classification.

4.3 Location Support

Citizens can enter or capture the location associated with the reported issue.

Browser geolocation functionality depends on browser support and user permission.

4.4 Priority Assessment

The system provides an AI-assisted priority assessment based on programmed rules and issue information.

This is intended to demonstrate how reports could be prioritized in a future production system.

4.5 Authority Recommendation

Based on the issue category, the system can recommend a relevant authority for the reported issue.

This recommendation is part of the prototype workflow and does not directly submit complaints to government authorities.

4.6 Multilingual Interaction

The platform is designed to support English, Tamil, and Sinhala user interaction.

Voice recognition functionality depends on browser support and user permissions.

4.7 Duplicate Detection

The backend checks submitted reports for potentially matching existing reports using relevant report information such as issue type and location. This helps demonstrate how repeated complaints could be identified.

4.8 Citizen Notifications

Citizens can receive notifications related to report activity and status changes.

4.9 Admin Alerts

The administrator interface supports monitoring of relevant report activity.

4.10 Report Tracking

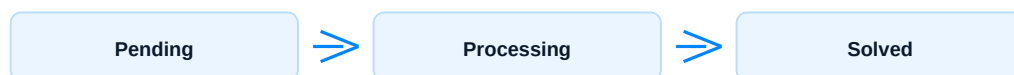
Citizens can view submitted reports and monitor their current status.

4.11 Resolution Verification

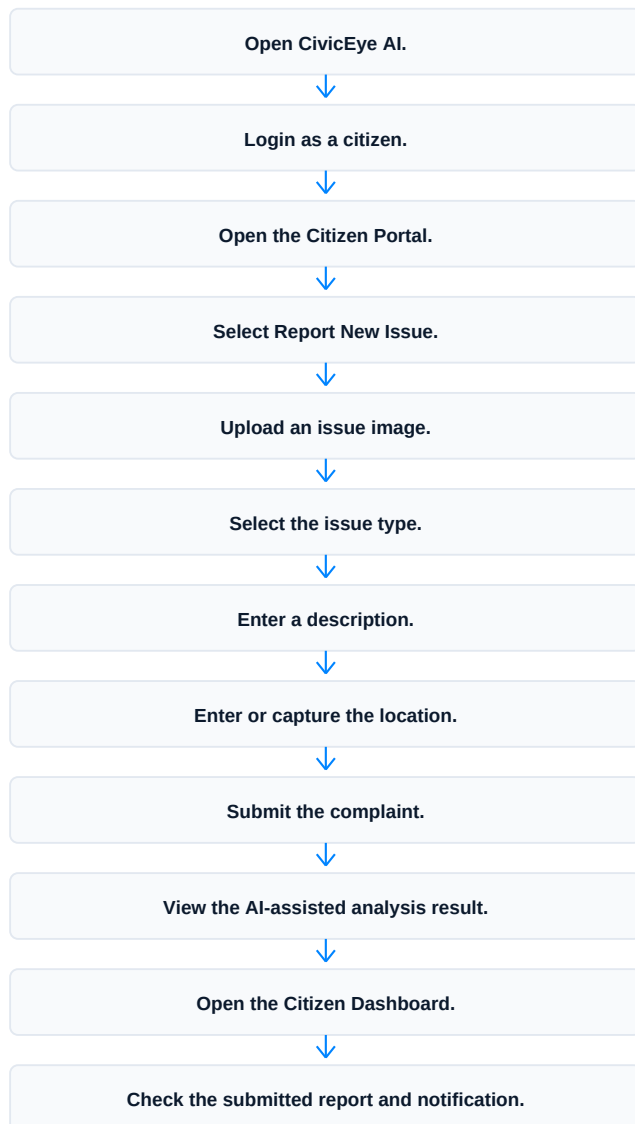
The prototype provides a citizen-facing resolution verification interaction after an issue is marked as solved.

4.12 Administrator Dashboard

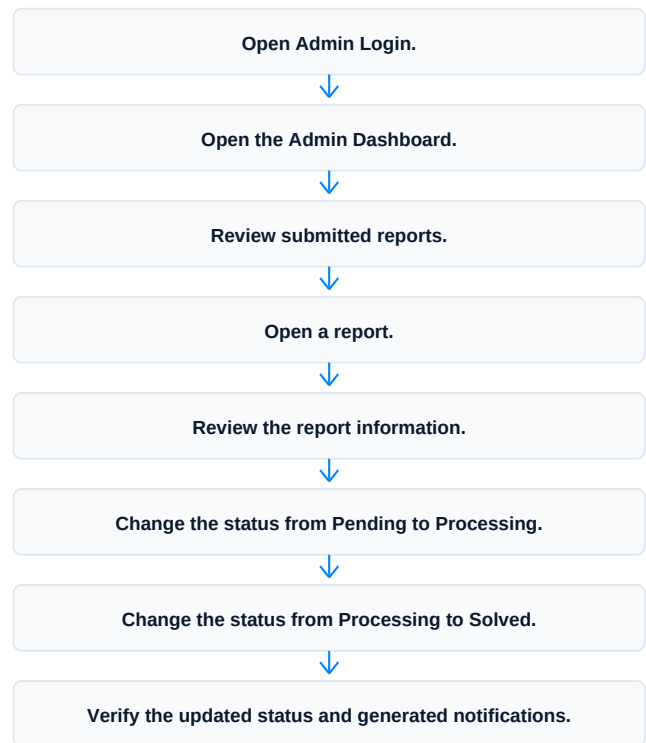
Administrators can review submitted reports and update their status.

EXAMPLE STATUS FLOW

Citizen Workflow



Admin Workflow



Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js
- CORS
- Multer
- dotenv

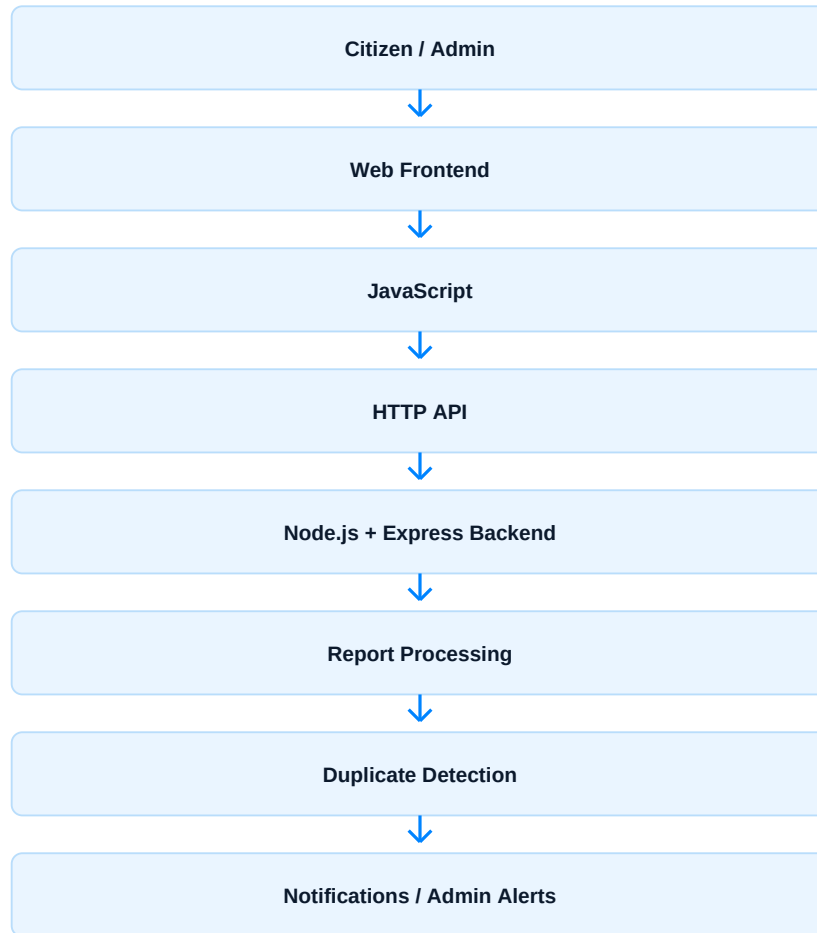
Development and Deployment

- Visual Studio Code
- Git / GitHub
- Vercel
- Railway

Browser-based capabilities

- Geolocation API
- Web Speech / Voice Recognition support where available

The CivicEye AI prototype follows a frontend-backend architecture.



The frontend communicates with the backend through HTTP API requests.

The backend handles report processing, duplicate detection, status management, notifications, and related application logic.

Main frontend pages include:

index.html

login.html

citizen.html

admin.html

report.html

ai-processing.html

track.html

profile.html

settings.html

notifications.html

privacy.html

Supporting frontend files include:

- script.js
- style.css
- images/

Backend files contain the Node.js and Express server, configuration, and supporting backend logic.

Frontend

The frontend can be deployed as a web application using Vercel.

Backend

The backend is deployed using Railway.

API Communication

The frontend communicates with the deployed backend through the configured backend API URL.

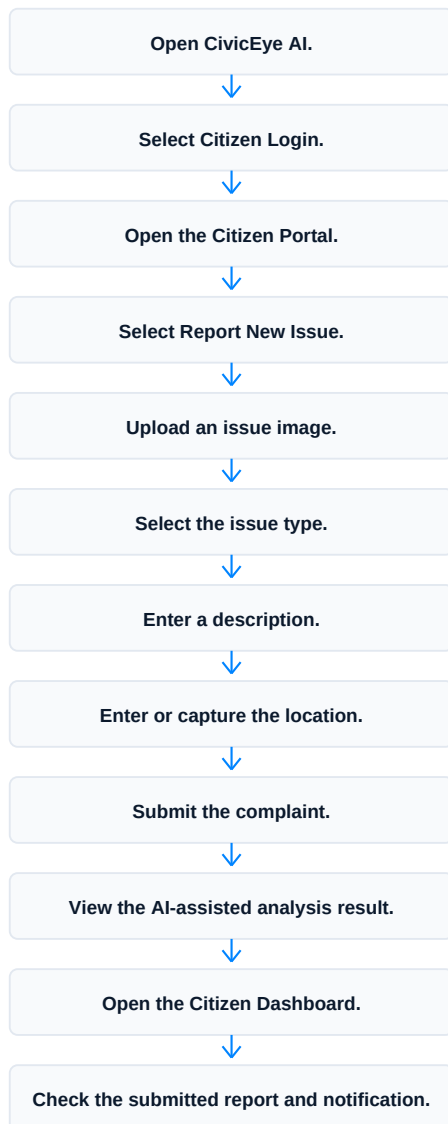
PORT

The backend uses the environment variable PORT when provided.

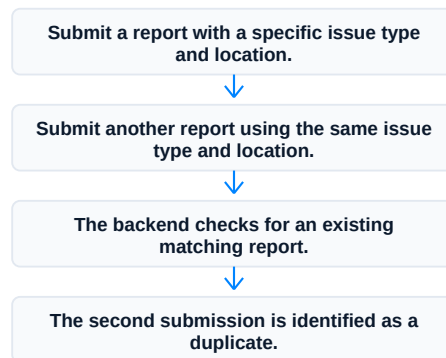
Railway

When deployed on Railway, the platform-provided port is used.

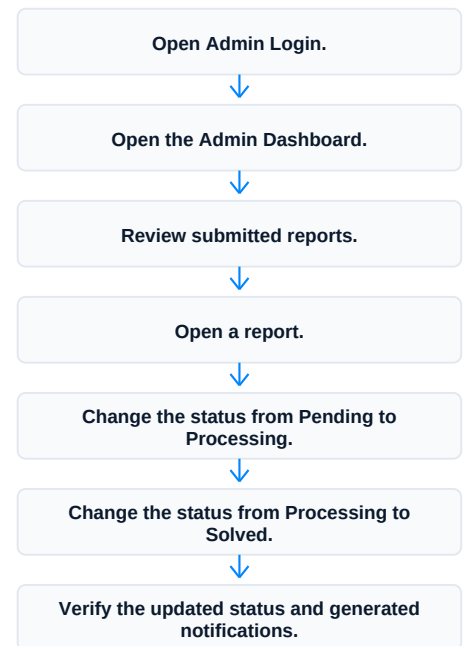
Citizen Report Test



Duplicate Detection Test



Admin Test



IMPORTANT PROTOTYPE LIMITATIONS

Rule-Based AI

The current AI-assisted analysis uses programmed rules and keyword matching. It does not currently use a trained machine-learning model or external AI API.

Confidence Display

The current prototype displays a predefined confidence value for demonstration purposes. It should not be interpreted as a statistically calibrated machine-learning confidence score.

Data Storage

Reports and notifications are currently stored in server memory. Therefore, the prototype does not provide permanent database storage.

Authentication

The current login system is designed for prototype demonstration and does not provide production-grade authentication.

Resolution Verification

The citizen resolution verification interface is currently a frontend prototype interaction and is not stored through a dedicated backend verification service.

Browser Features

Voice recognition and geolocation depend on browser support and user permissions.

FUTURE IMPROVEMENTS

- | | |
|--------------------------------------|--|
| ● Trained machine-learning models | ● Computer vision for image-based issue classification |
| ● Permanent cloud database | ● Secure user authentication |
| ● Role-based access control | ● Per-citizen report access |
| ● Real-time authority communication | ● Advanced GIS mapping |
| ● Persistent resolution verification | ● Advanced analytics and reporting |
| ● Mobile application support | ● Integration with relevant government authorities |

PROJECT OBJECTIVE

The objective of CivicEye AI is to demonstrate how web technologies and AI-assisted decision logic can support more structured civic issue reporting and monitoring.

The project focuses on:

- Simplifying citizen reporting
- Organizing complaint information
- Supporting issue prioritization
- Reducing repeated complaints
- Recommending relevant authorities
- Improving complaint visibility
- Supporting multilingual interaction

CONCLUSION

CivicEye AI demonstrates a prototype workflow for digital civic issue reporting and monitoring.

By combining a citizen-facing web interface, a Node.js backend, rule-based AI-assisted analysis, duplicate detection, authority recommendation, notifications, and an administrator dashboard, the project provides a foundation for a future intelligent civic management platform.

The current implementation is intentionally presented as a prototype, with future scope for machine learning, persistent databases, secure authentication, advanced mapping, and direct authority integration.